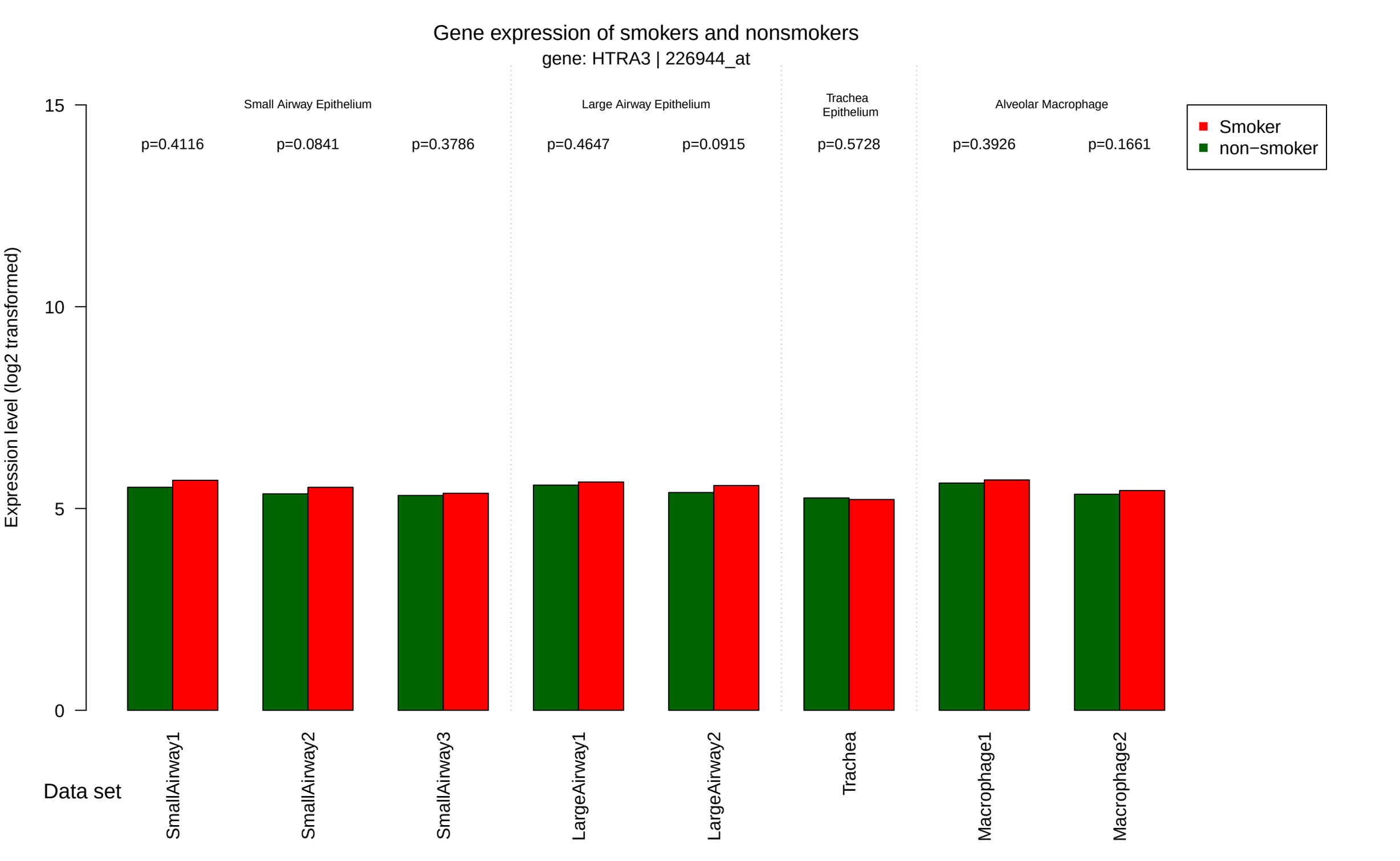
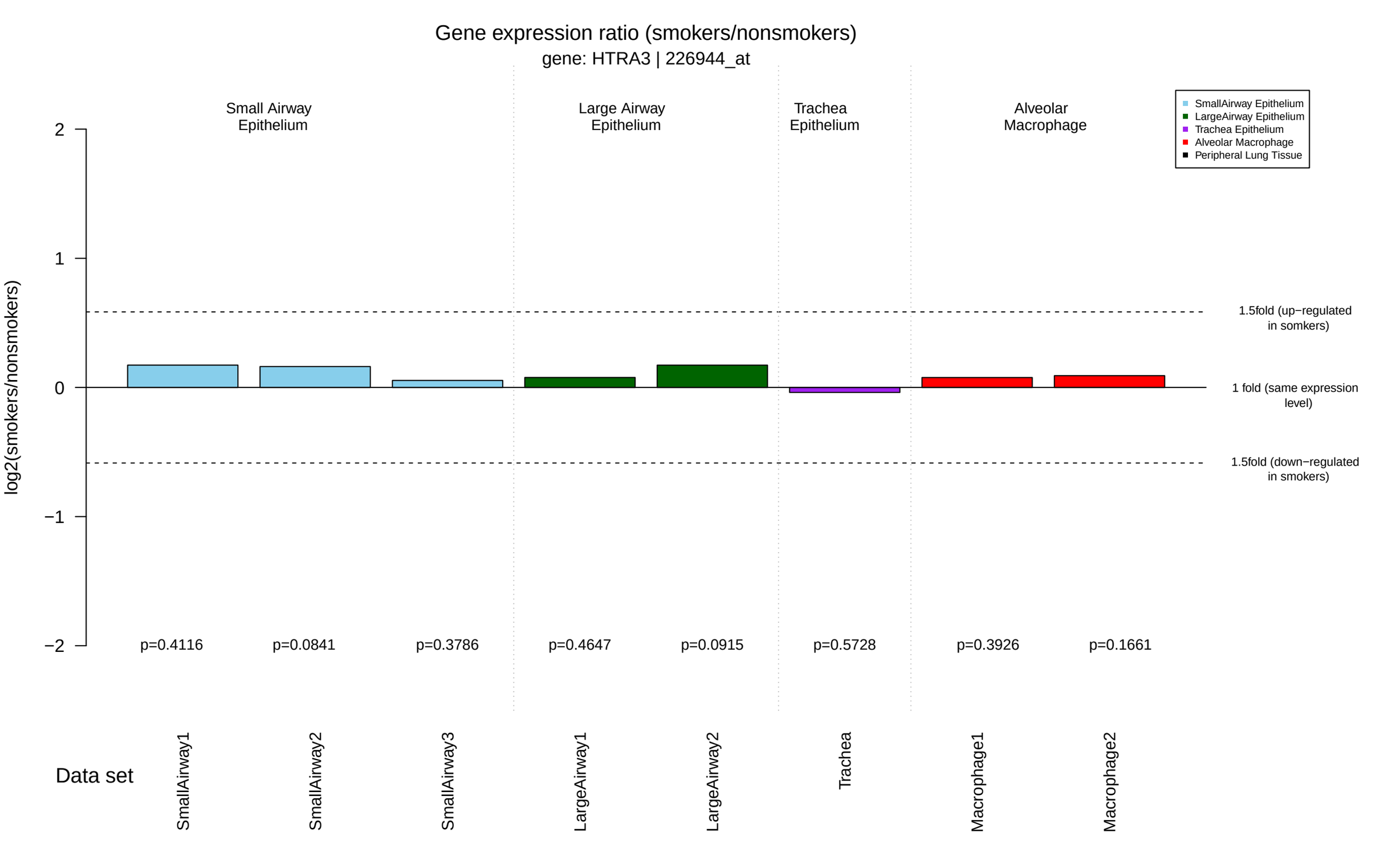
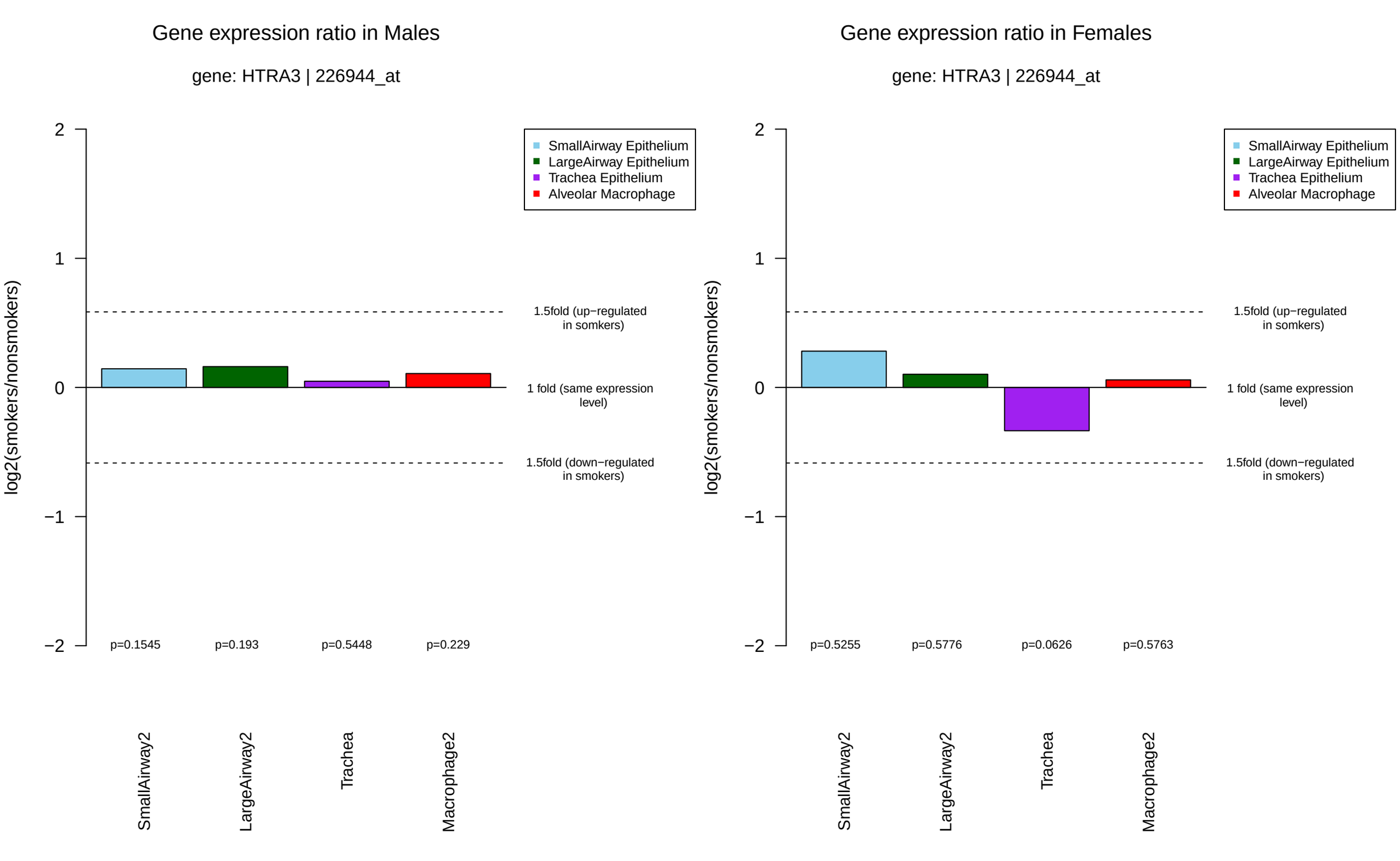
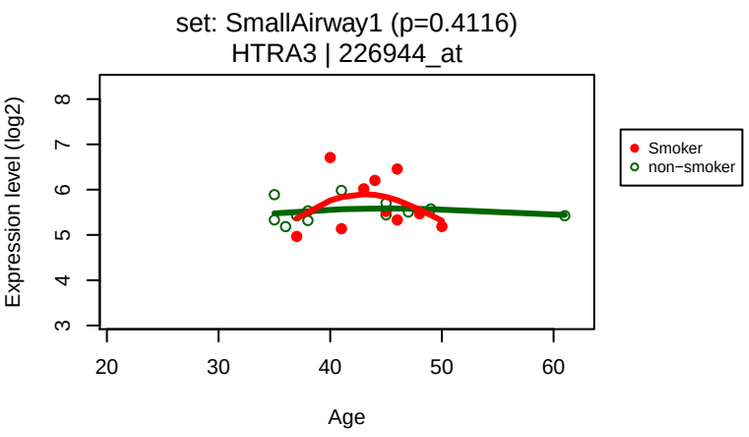
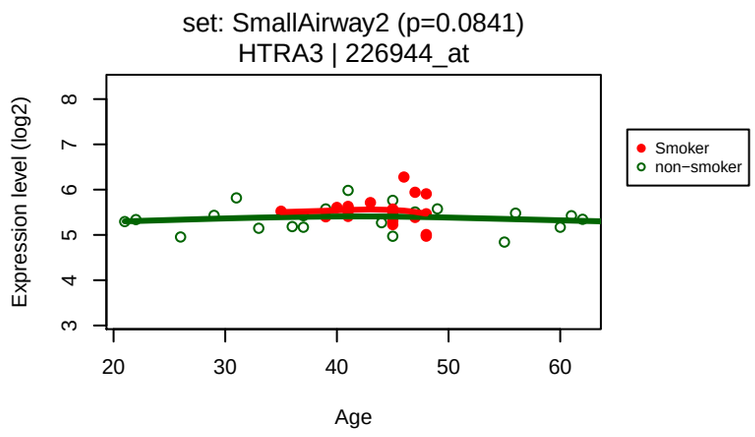
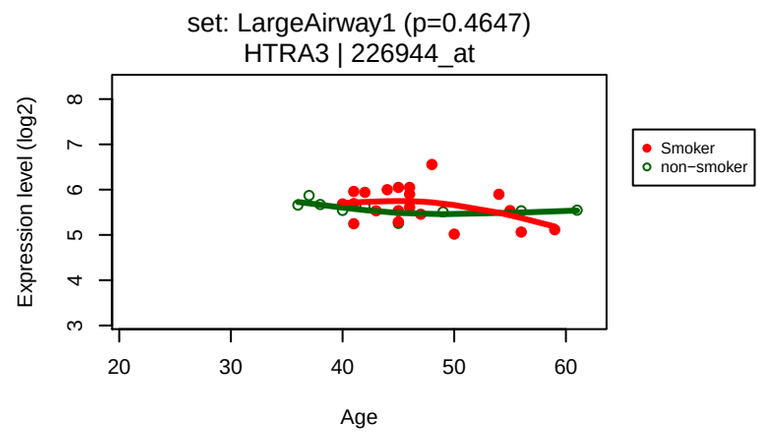

Gene expression of smokers and nonsmokers
gene: HTRA3 | 226944_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4116
p=0.0841
p=0.3786
p=0.4647
p=0.0915
p=0.5728
p=0.3926
p=0.1661
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

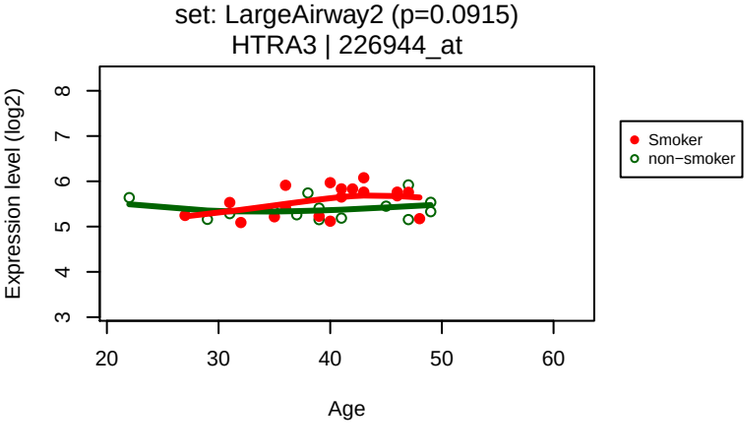

Gene expression ratio (smokers/nonsmokers)
gene: HTRA3 | 226944_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4116
p=0.0841
p=0.3786
p=0.4647
p=0.0915
p=0.5728
p=0.3926
p=0.1661
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

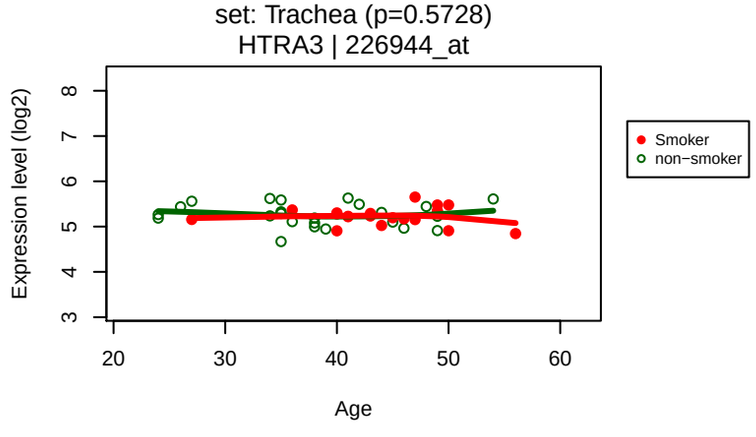

Gene expression ratio in Males
gene: HTRA3 | 226944_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1545
p=0.193
p=0.5448
p=0.229
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HTRA3 | 226944_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5255
p=0.5776
p=0.0626
p=0.5763
SmallAirway2
LargeAirway2
Trachea
Macrophage2

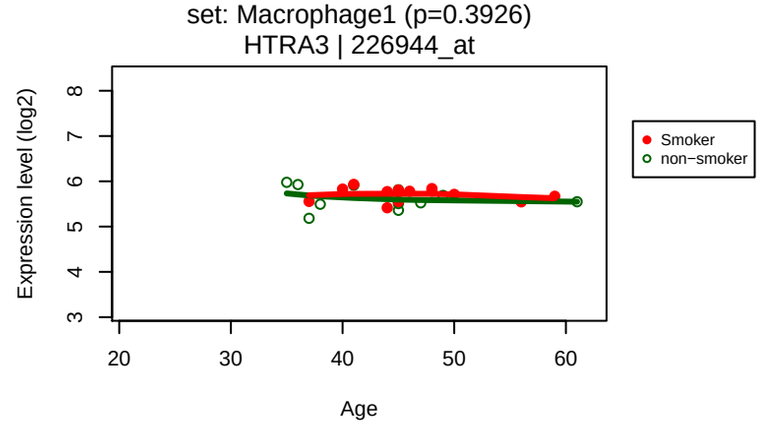

set: SmallAirway1 (p=0.4116)
HTRA3 | 226944_at
Expression level (log2)
Age
Smoker
non−smoker

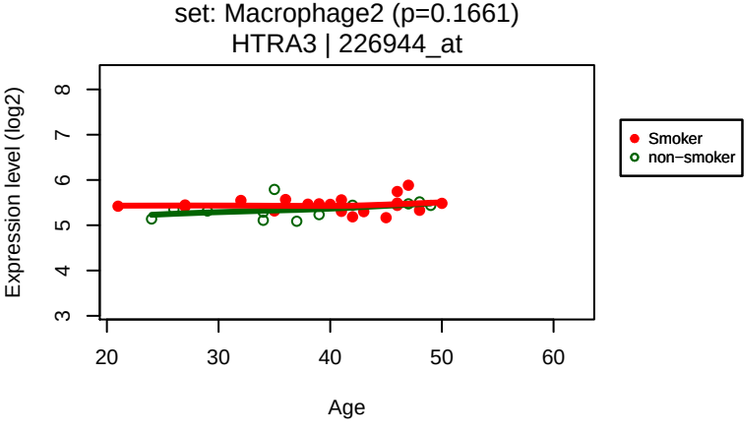

set: SmallAirway2 (p=0.0841)
HTRA3 | 226944_at
Expression level (log2)
Age
Smoker
non−smoker

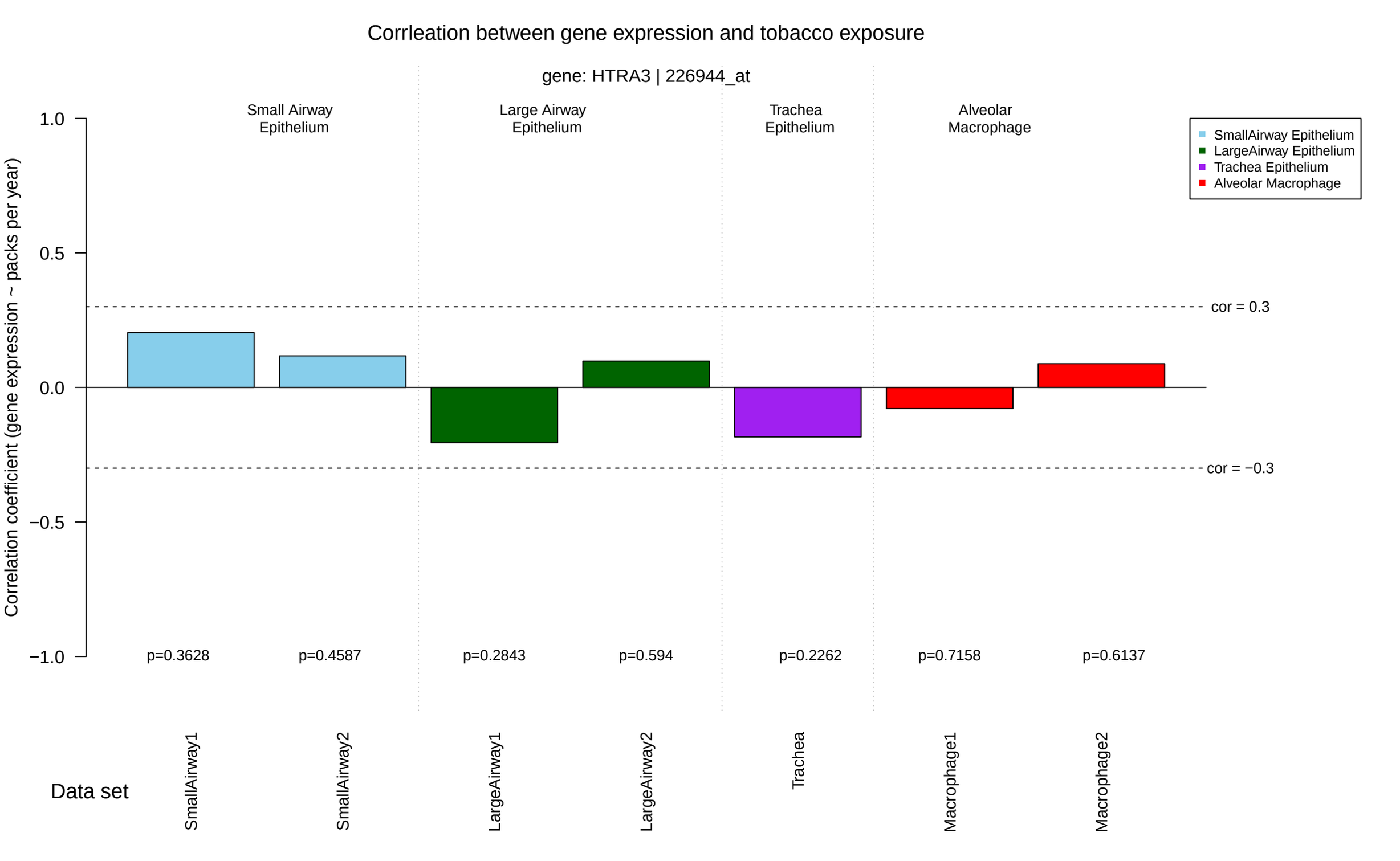

set: LargeAirway1 (p=0.4647)
HTRA3 | 226944_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0915)
HTRA3 | 226944_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5728)
HTRA3 | 226944_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3926)
HTRA3 | 226944_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1661)
HTRA3 | 226944_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HTRA3 | 226944_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3628
p=0.4587
p=0.2843
p=0.594
p=0.2262
p=0.7158
p=0.6137
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2